

# SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

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## COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

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QUESTIONS: 2

Duration : 1 Hour  
Total Marks:30

Annexure A

## Constraint-Based Problem Solving

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Code of Conduct:

I, Vinod Kumar [19232567] (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

[Signature]  
Signature of the student:

2022-23

## CO2-A12. AI

A workshop must assign 3 technicians (Tech 1, Tech 2, Tech 3) to repair 3 machines (Machine X, Machine Y, Machine Z) such that:

Constraints:-

- i, Tech 1 does not have the skill to repair Machine Z.
- ii, Tech 2 must be assigned before Tech 3 (Machine X < Machine Y < Machine Z)
- iii, Only one technician per machine.
- iv, Tech 3 cannot repair Machine X
- v, All technicians must be assigned exactly one machine.

Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

Problem Formulation:-

- Variables:-  $M_x, M_y, M_z$  representing Machine X, Machine Y, Machine Z.
- Domains:- For each machine variable, domain  $D = \{T_1, T_2, T_3\}$  where,  $T_i$  denotes technician  $i$ .
- Constraints:-
  - 1, Skill Constraint:-  $M_z \neq T_1$  (Tech 1 cannot repair Machine Z).
  - 2, Precedence Constraint:- Tech 2 must be assigned to a machine before Tech 3 across the machine sequence.  $(M_x, M_y, M_z)$ . Formally,  $Pos(T_2) < Pos(T_3)$ .



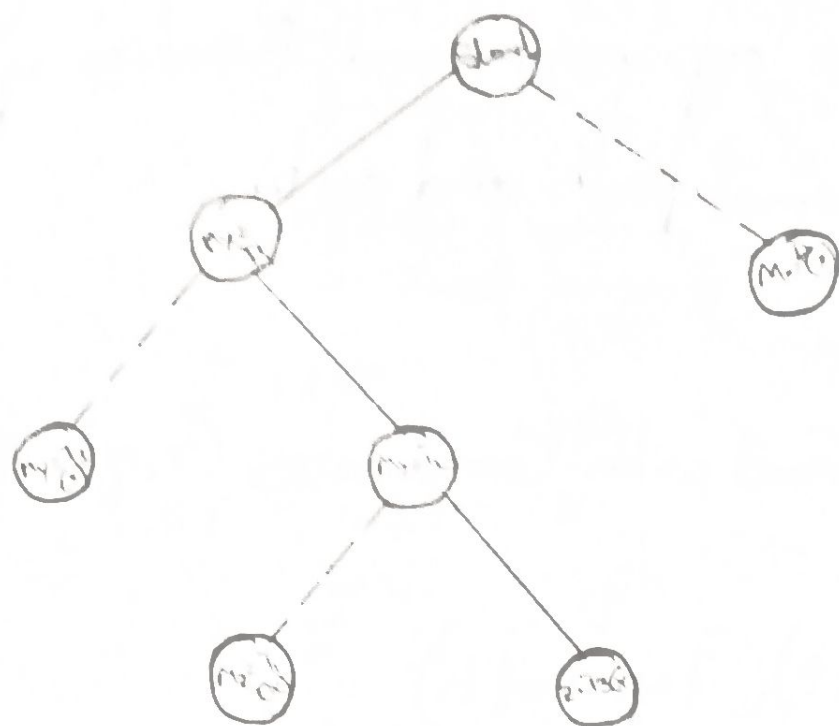
3. Bijection/Uniqueness Constraint:- All  
 incl:  $M_x \neq M_y, M_y \neq M_z, M_x \neq M_z$ .
4. Skill Constraint:  $M_x \neq T_3$  (Tech 3 cannot repair Machine 1)
5. Complete Assignment:- All 3 technicians must be assigned exactly 1 machine.

### Step-by-Step Backtracking Search Execution:

We perform standard Depth-First Backtracking Search with chronological variable order ( $M_x$  1st,  $M_y$  2nd,  $M_z$  3rd) and domain value order ( $T_1$  1st,  $T_2$  2nd,  $T_3$  3rd).

| Step | Assignment Choice | Constraint Check                                                                                        | Status & Reasoning                        |
|------|-------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 1    | $M_x = T_1$       | $M_x \neq T_3$                                                                                          | VALID proceed to $M_y$                    |
| 2    | $M_y = T_1$       | $M_x \neq M_y$                                                                                          | FAIL $T_1$ already assigned to $M_x$ .    |
| 3    | $M_y = T_2$       | $M_x \neq M_y$                                                                                          | VALID Proceed to $M_z$ .                  |
| 4    | $M_z = T_1$       | $M_z \neq T_1$                                                                                          | FAIL Violates Constraint (1)              |
| 5    | $M_z = T_2$       | $M_x \neq M_z$                                                                                          | FAIL $T_2$ already assigned to $M_y$ .    |
| 6    | $M_z = T_3$       | 1. $M_z \neq T_1$ (Pass)<br>2. All distinct (Pass)<br>3. $Pos(T_2) < Pos(T_3)$<br>( $M_y < M_z$ , Pass) | SUCCESS Complete & valid Assignment Found |

## Branch Space Tree Diagram!:-



### Final Valid Schedule Result:-

- Machine X : Technician 1 (Tech 1)
- Machine Y : Technician 1 (Tech 2)
- Machine Z : Technician 3 (Tech 3).

Verification!:- Tech 1 is on Machine X (Not Z - Satisfied)  
Tech 2 is on Machine Y (before Tech 3 on Machine Z - Satisfied)  
Tech 3 is on Machine Z (Not X - Satisfied). All technicians assigned to 1 machine (Satisfied).

2) A delivery robot operates in a grid environment (5x5). It must move from Start (S) to Goal (G). Obstacles block certain cells.

### Constraints:-

- i, Movement allowed: Up, Down, Left, Right.
- ii, Cost of each move = 1
- iii, Cannot pass through obstacles.



iv, Use Manhattan Distance heuristic.  
Using the above constraints and BFS Search Algorithm,  
show step-by-step node expansion and frontier updates,  
and determine the optimal path and cost.

#### A) Environment & Setup:-

- Grid Domain:-  $5 \times 5$  Grid with Coordinates  $(x, y)$  where  $x, y \in \{0, 1, 2, 3, 4\}$ .
- Start State( $S$ ):-  $(0, 0)$  (Top-Left).
- Goal State( $G$ ):-  $(4, 4)$  (Bottom-Right).
- Obstacles( $O$ ):-  $(1, 1), (1, 2), (2, 1), (3, 3)$ .
- Movement Actions:- Up, Down, Left, Right (Unit Step Cost = 1).
- Heuristic Evaluation (Manhattan Distance):  $h(x, y) = |x - 4| + |y - 4|$ .

Note:- Standard Breadth-First Search expands strictly level-by-level using a FIFO queue structure, guaranteeing optimal path length in unweighted graphs.



# 5 Grid Visualization & Manhattan Distance Values:

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|              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|
| S<br>h=8     | (0,1)<br>h=7 | (0,2)<br>h=6 | (0,3)<br>h=5 | (0,4)<br>h=4 |
| (1,0)<br>h=7 | OBS          | OBS          | (1,3)<br>h=4 | (1,4)<br>h=3 |
| (2,0)<br>h=6 | OBS          | (2,2)<br>h=4 | (2,3)<br>h=3 | (2,4)<br>h=2 |
| (3,0)<br>h=5 | (3,1)<br>h=4 | (3,2)<br>h=3 | OBS          | (3,4)<br>h=1 |
| (4,0)<br>h=4 | (4,1)<br>h=3 | (4,2)<br>h=2 | (4,3)<br>h=1 | G<br>h=0     |

## Step-by-step BFS Expansion Trace:-

| Step | Current Node | Action/Neighbors Generated        | Frontier (FIFO)       | Visited Set.          |
|------|--------------|-----------------------------------|-----------------------|-----------------------|
| 0    | -            | Initialize                        | [(0,0)]               | {(0,0)}               |
| 1    | (0,0)        | Down: (1,0), Right: (0,1)         | [(1,0), (0,1)]        | {(0,0), (1,0), (0,1)} |
| 2    | (1,0)        | Down: (2,0) [Right blocked]       | [(0,1), (2,0)]        | {..., (2,0)}          |
| 3    | (0,1)        | Right: (0,2) [Down blocked]       | [(2,0), (0,2)]        | {..., (0,2)}          |
| 4    | (2,0)        | Down: (3,0) [Right blocked]       | [(0,2), (3,0)]        | {..., (3,0)}          |
| 5    | (0,2)        | Right: (0,3) [Down blocked]       | [(3,0), (0,3)]        | {..., (0,3)}          |
| 6    | (3,0)        | Down: (4,0), Right: (3,1)         | [(0,3), (4,0), (3,1)] | {..., (4,0), (3,1)}   |
| ...  | ...          | (Level expansions continue) ..... | .....                 | .....                 |
| N    | (4,3)        | Right: (4,4) [Goal Reached]       | [(4,4), ...]          | {..., (4,4)}          |

## Optimal Path & Cost Summary :-

- Optimal Path Sequence:  $(0,0)$   $\rightarrow$   $(1,0)$   $\rightarrow$   $(2,0)$   $\rightarrow$   $(3,0)$   $\rightarrow$   $(3,1)$   $\rightarrow$   $(3,2)$   $\rightarrow$   $(4,2)$   $\rightarrow$   $(4,3)$   $\rightarrow$   $(4,4)$ .
- Total path Cost (Total Steps) :- 8 units.
- Heuristic Accuracy :- Initial Manhattan distance  $h(0,0) = |0-4| + |0-4| = 8$ . Because the robot navigates efficiently around the obstacle block, the optimal cost matches the theoretical lower bound.